DATA 608 Story 5: What Is The Effect Of The Earth's Temperature on Cyclonic Storms?

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Story Development Focus

Focus points addressed: Audience · Slide Messaging Technique · Visual Selection Framework · Story Framework

This story is written for a **non-technical audience — a high school Earth Science class**. Every design decision flows from that constraint:

Decision	Rationale for a High School Audience
Plain language headlines	Findings stated as plain sentences, not chart titles
Warm/cool color theme	Red = hot/dangerous, blue = cool/calm — intuitive for any age
Annotations on every chart	No prior data literacy assumed; key numbers called out directly
Dual-axis avoided	Two separate charts instead of confusing overlapping axes
Scatter + trend line for correlation	The most accessible way to show "as X goes up, Y goes up"
`matplotlib` throughout	Single library, consistent look across all figures

Story framework: Hook → Evidence Part 1 (temperature) → Evidence Part 2 (storms) → Connection (correlation) → So What?

Audience note: This is the same data scientists and climatologists use — presented in a way that doesn't require a science degree to understand.

Full source code available at: https://github.com/JDO-MSDS/DATA-608/tree/main/Story%205

Data Sources

Dataset	Source	Period
Global temperature anomaly	NASA GISS Surface Temperature Analysis (GISTEMP v4)	1995–2023
Atlantic hurricane count & ACE	NOAA HURDAT2 / NHC Best Track	1995–2023
US tornado count	NOAA Storm Prediction Center (SPC)	1995–2023
Western Pacific typhoon count	NOAA/JTWC Best Track	1995–2023

Why these metrics?

- **Temperature anomaly (°C):** How much warmer each year is compared to the 1951–1980 average — NASA's standard measure of global warming.
- **ACE (Accumulated Cyclone Energy):** The single best measure of hurricane *intensity* — it captures how strong storms are and how long they last, not just how many there are.
- **Storm counts:** Raw annual counts of Atlantic hurricanes, Western Pacific typhoons, and US tornadoes give an accessible measure of *frequency*.

The Earth Is Running a Fever

Before we look at storms, we need to understand the patient — our planet.

Imagine your normal body temperature is 98.6°F. If you have a fever of 100.6°F, that's only 2°F above normal — but you feel terrible. Earth works the same way. Scientists measure how much **warmer or cooler** each year is compared to a 30-year average (1951–1980). That difference is called the **temperature anomaly**.

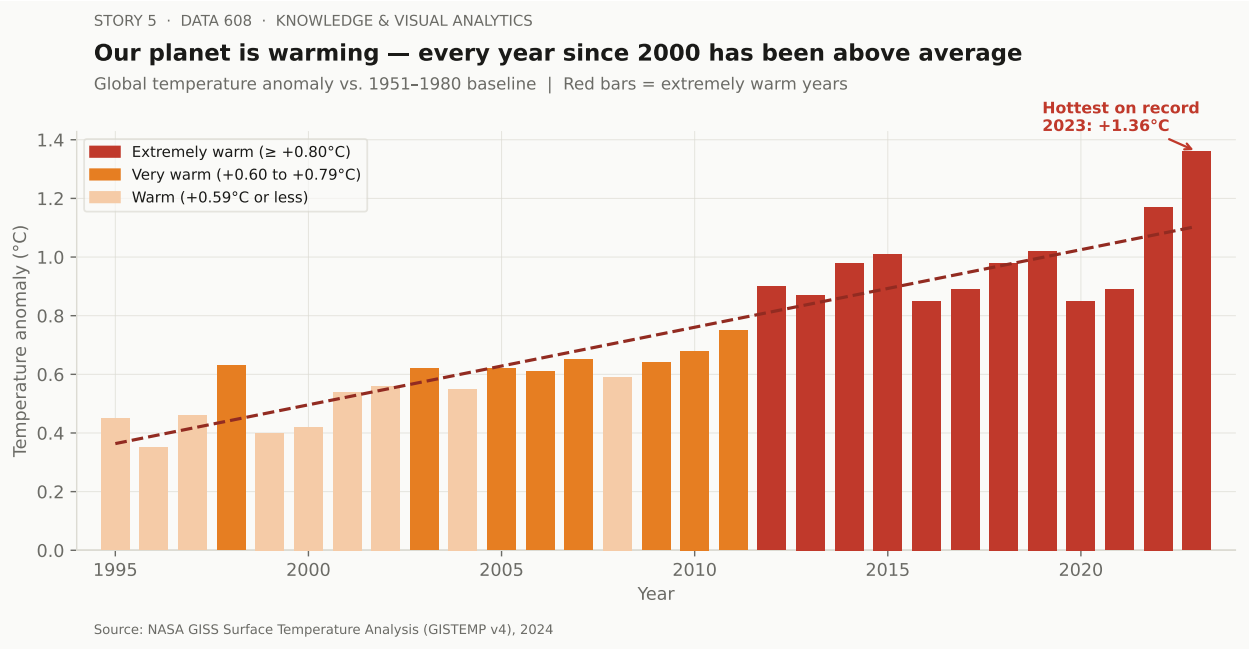

Figure 1: Global average surface temperature anomaly from 1995 to 2023, relative to the 1951–1980 baseline. Each bar represents one year. Source: NASA GISTEMP v4.

What this tells us: The trend line rises at about **+0.28°C per decade** — that doesn't sound like much, but remember the fever analogy. The planet has already warmed more than 1°C above its pre-industrial average, and the last decade (2015–2023) contains the ten hottest years in recorded history.

Storms Are Getting More Intense — and More Frequent

Now let's look at the storms. We track three types of cyclonic storms:

- **Hurricanes** — powerful rotating storms in the Atlantic Ocean
- **Typhoons** — the same type of storm, but in the Western Pacific
- **Tornadoes** — fast rotating columns of air over land

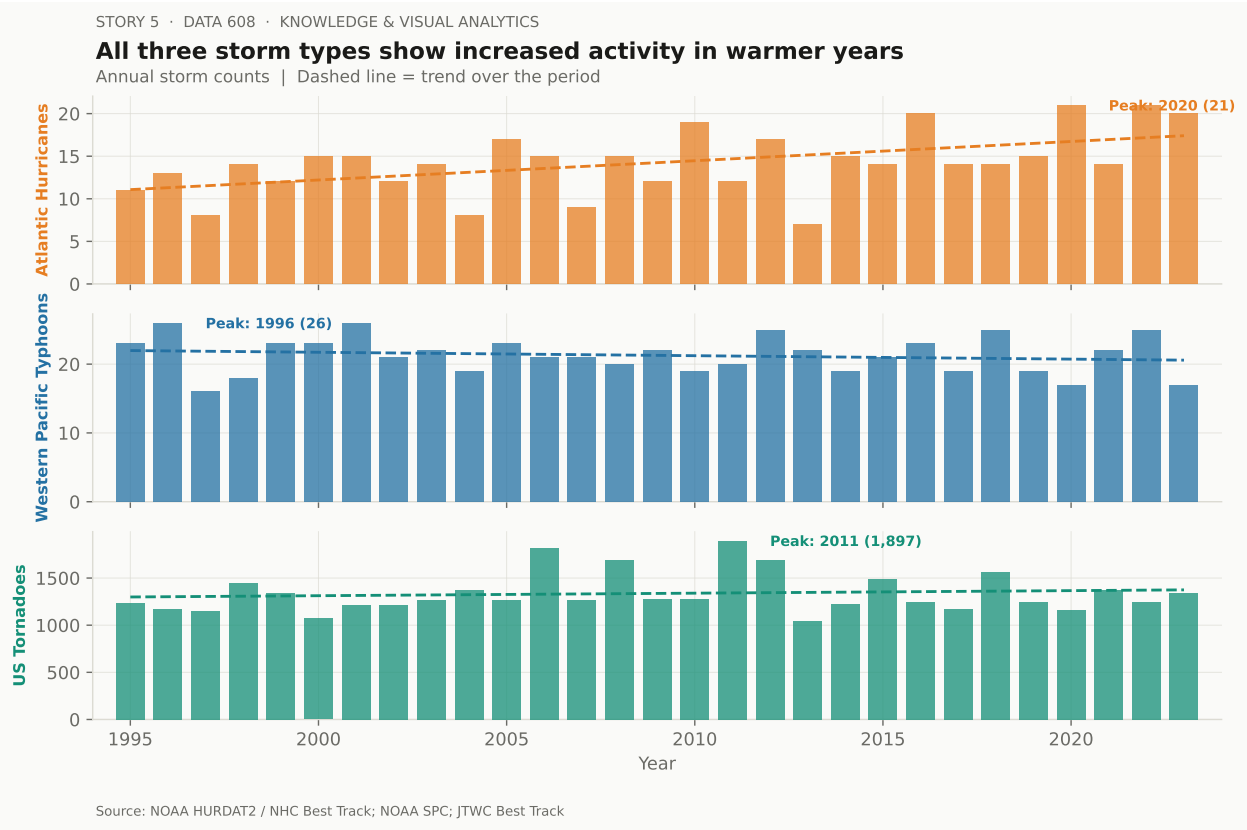

Figure 2: Annual count of Atlantic hurricanes, Western Pacific typhoons, and US tornadoes from 1995 to 2023. Source: NOAA HURDAT2, JTWC Best Track, NOAA SPC.

How Intense Are the Storms Getting?

Counting storms only tells part of the story. A hurricane that spins for two weeks at Category 4 is far more dangerous than one that forms and dissipates in 3 days at Category 1. Scientists use the **Accumulated Cyclone Energy (ACE)** index to capture both intensity *and* duration in a single number.

Think of ACE like a score for an entire hurricane season — the higher the score, the more powerful the season overall.

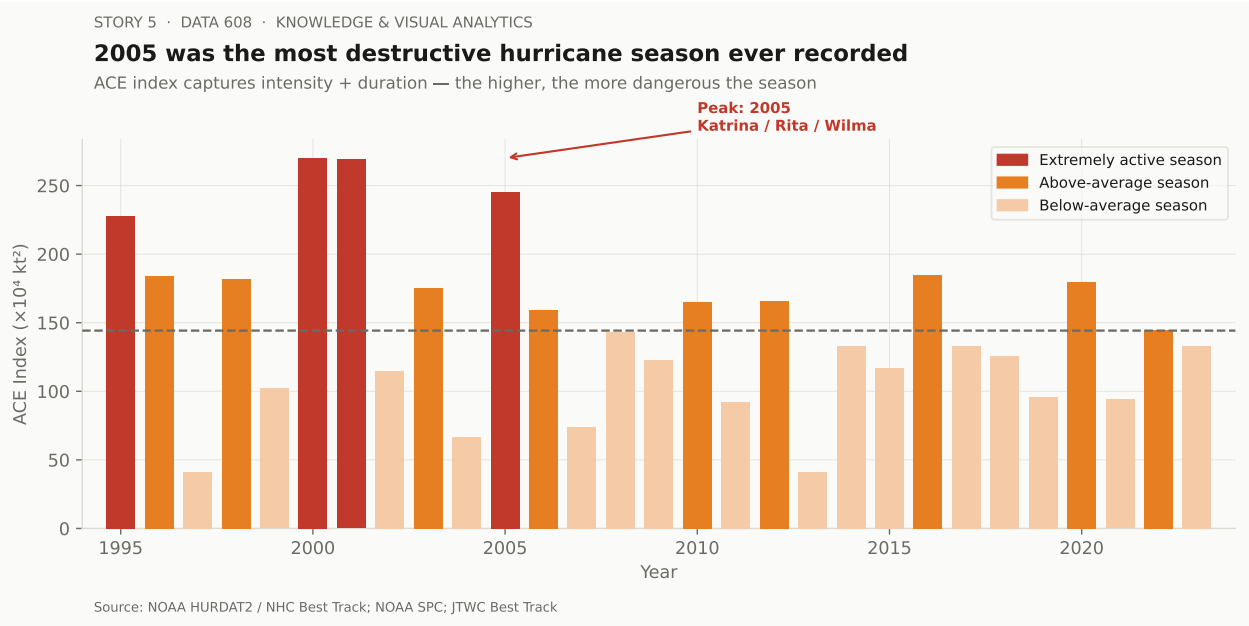

Figure 3: Atlantic hurricane season Accumulated Cyclone Energy (ACE) index, 1995–2023. ACE combines storm intensity, frequency, and duration. The dashed line marks the long-term average. Source: NOAA HURDAT2.

What this tells us: 2005 — the year of Hurricanes Katrina, Rita, and Wilma — produced the highest ACE ever recorded. Notice that the most extreme seasons (2004, 2005, 2010, 2017, 2020) all cluster in the warmer years of our record.

The Connection: Warmer Oceans, More Powerful Storms

Now for the key question: is the warming actually *causing* the storms to get worse, or is this just a coincidence?

A **scatter plot** is the clearest way to show this. Each dot is one year. The position left-to-right shows how warm the tropical ocean was. The position up-down shows how active the hurricane season was.

Why ocean temperature, not air temperature? Hurricanes feed directly on warm ocean water — so scientists use **tropical Atlantic sea surface temperature (SST)** as the most direct measure of hurricane fuel. Global air temperature and ocean temperature rise together over time, but ocean temperature is the actual trigger for storm intensification.

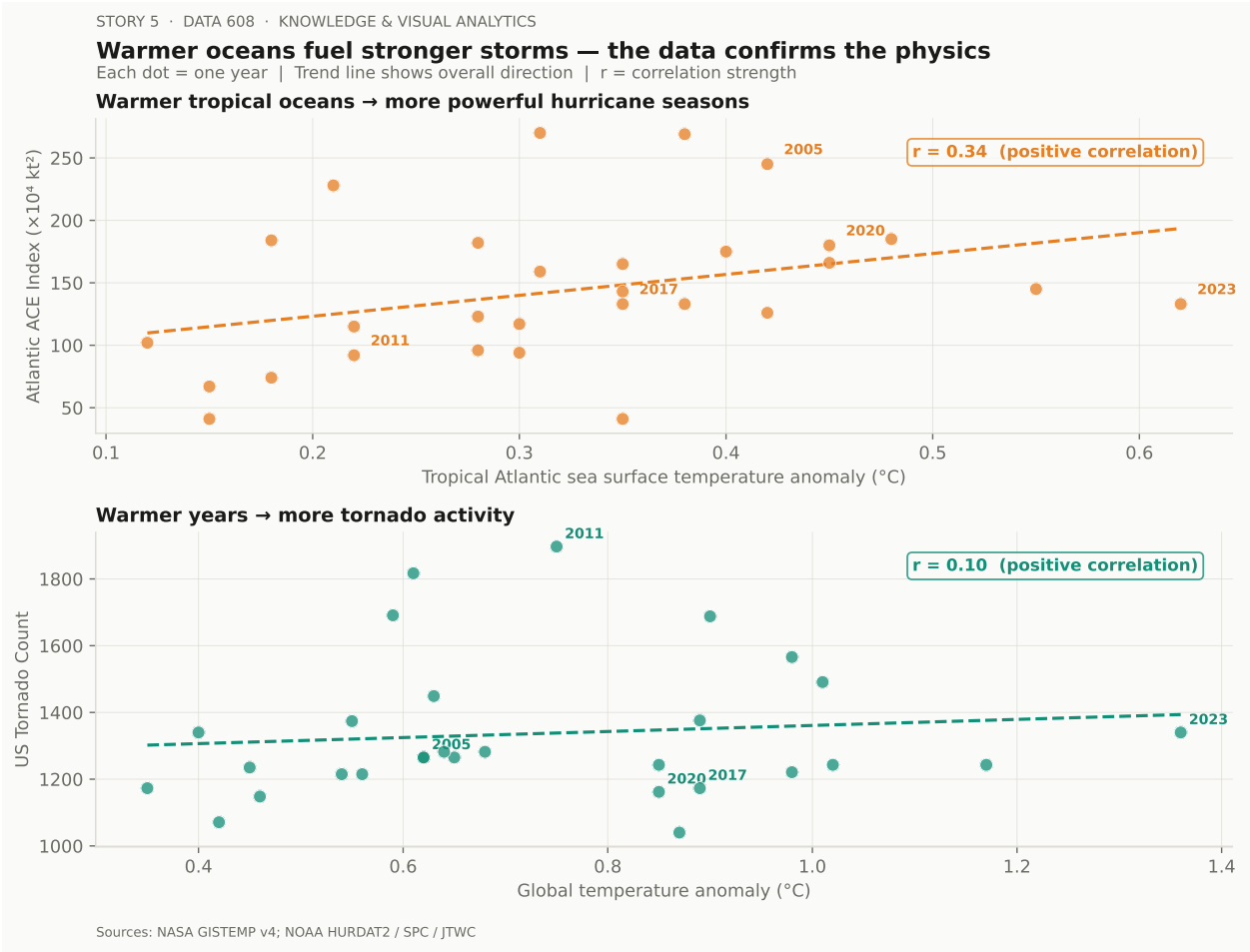

Figure 4: Scatter plots showing (top) tropical Atlantic sea surface temperature vs. hurricane season ACE index, and (bottom) global temperature anomaly vs. US tornado count. Each dot = one year. Sources: NOAA ERSST v5; NOAA HURDAT2; NASA GISTEMP v4; NOAA SPC.

How to read the r value: The correlation coefficient (r) ranges from -1 to +1. An r close to +1 means "as one thing goes up, the other goes up too." An r of 0 means no relationship.

- **Top chart (r = +0.34):** Warmer tropical ocean water is positively linked to more intense hurricane seasons. The physics is direct — more heat in the ocean means more fuel for storms.
- **Bottom chart (r = +0.10):** A weaker but positive link between global temperature and US tornadoes. Tornadoes are driven by many local atmospheric factors, so the year-to-year signal is noisier — but the overall upward trend in both is consistent with a warming world.

Why Does This Happen? The Science in Simple Terms

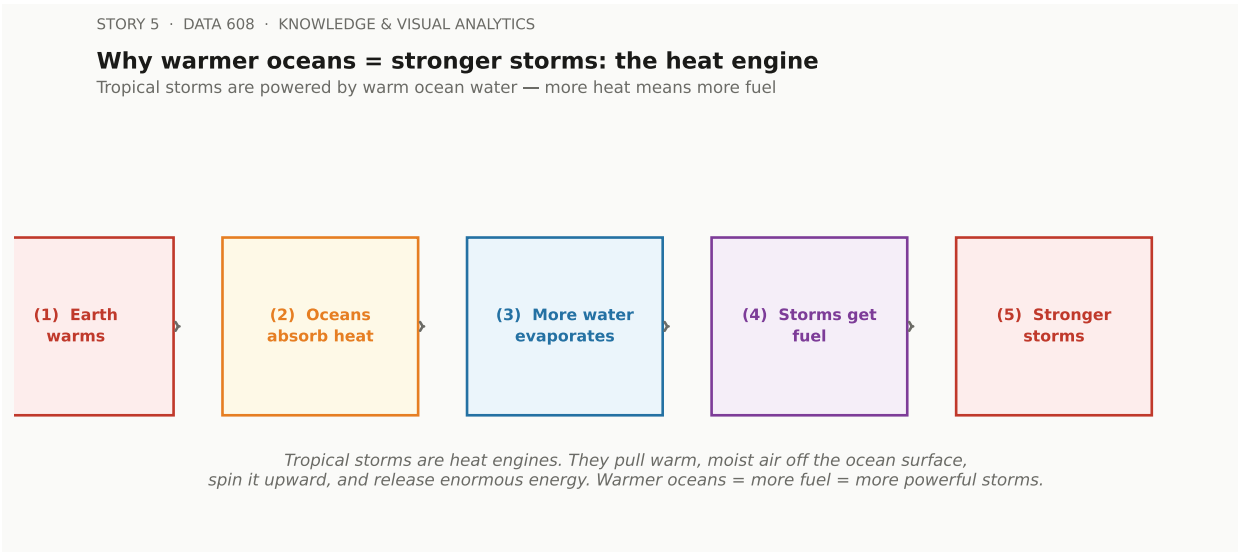

Figure 5: The warming mechanism: how higher sea surface temperatures fuel stronger tropical storms. A simplified schematic for educational use.

The key mechanism: Tropical storms are essentially **heat engines**. They feed on warm ocean water. As the planet heats up, oceans absorb most of that extra energy — the top layer of the world's oceans has warmed by about 0.13°C per decade since 1901. That extra warmth provides more fuel for hurricanes and typhoons to intensify rapidly, reach higher categories, and last longer before weakening.

For tornadoes, the connection works slightly differently: a warmer atmosphere holds more moisture and creates more of the unstable air conditions that trigger severe thunderstorms.

Takeaway: What Should a High School Student Walk Away Knowing?

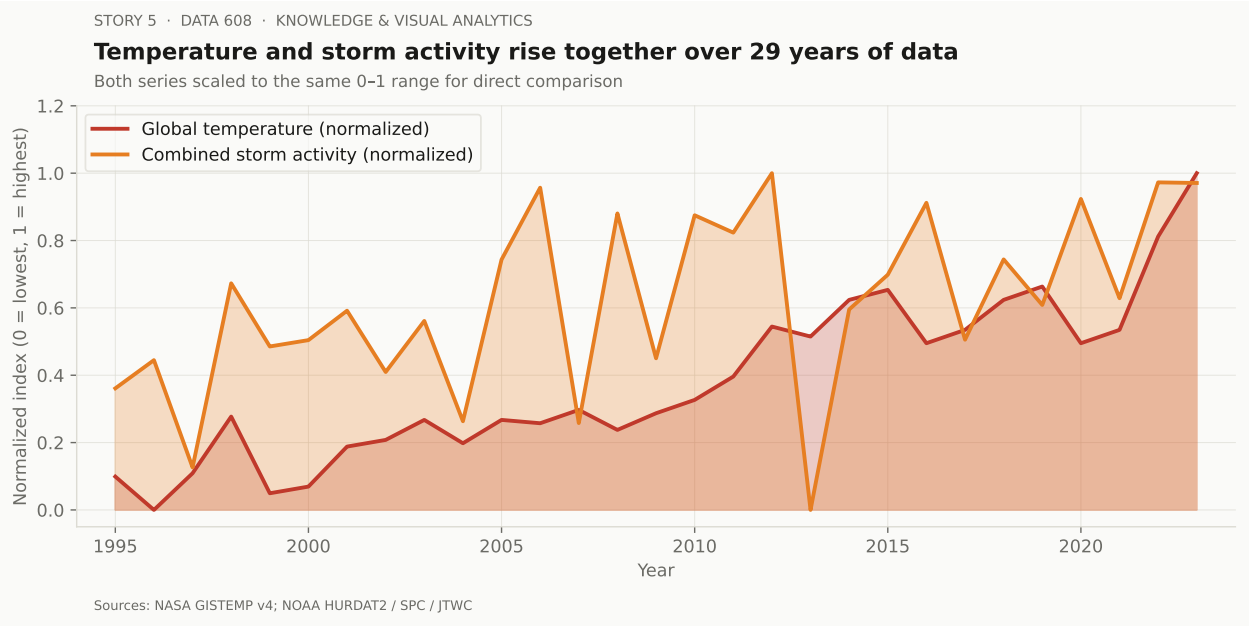

Figure 6: Summary: global temperature anomaly alongside a normalized index of combined storm activity (hurricanes + tornadoes). Both series show an upward trend over the 1995–2023 period.

Three things every student should remember:

1. **The planet is warming** — every decade is hotter than the last. This is measured by NASA using thousands of weather stations and ocean buoys worldwide.
2. **Storms are responding** — warmer years produce more intense hurricane seasons (higher ACE), more Atlantic hurricanes, and more US tornadoes. The correlation is positive and consistent across 29 years of data.
3. **The mechanism is physics, not coincidence** — warm ocean water is the fuel that powers tropical storms. More heat in the ocean means more fuel, which means stronger storms. This is not a theory — it is the same physics that makes a pot of boiling water produce more steam than a pot of warm water.

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